THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 8	Diameter $= 2\times$ radius.
2	**(b)** 5	Radius $=$ diameter $\div 2$.
3	**(a)** a compass	Property
4	**(c)** 9	Opposite sides of a rectangle are equal.
5	**(c)** 90	All angles of a square are right angles.
6	**(b)** 360	4 right angles.
7	**(d)** 14	Diameter $= 2\times$ radius.
8	**(a)** 8	Radius $=$ diameter $\div 2$.
9	**(a)** chord	Property
10	**(a)** 12	Opposite sides of a rectangle are equal.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diameter $= 2 \times 6 = 12$ cm (1); Longest chord is the diameter, 12 cm (1).	1+1
12	A rectangle (1); Other sides are 8 cm and 5 cm (1).	1+1
13	All sides are 7 cm (1); Perimeter $= 4 \times 7 = 28$ cm (1).	1+1
14	Diameter $= 2 \times 9 = 18$ cm (1); Longest chord is the diameter, 18 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius $= 5$ cm (1); Chord 7 cm vs diameter 10 cm (1); No: the diameter is the longest chord (1).	1+1+1
16	Open the compass to 3 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 3 cm from O (1).	1+1+1
17	Perimeter $= 2(9 + 4) = 26$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1
	OR: Radius $= 8$ cm (1); Chord 15 cm vs diameter 16 cm (1); No: the diameter is the longest chord (1).	1+1+1
18	Open the compass to 4 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 4 cm from O (1).	1+1+1
	OR: Perimeter $= 2(12 + 8) = 40$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 6 cm (1); Draw perpendicular at B and mark BC = 4 cm (1); Draw perpendicular at A and mark AD = 4 cm, join D and C (1); Check that DC = 6 cm (1); Perimeter $= 2(6 + 4) = 20$ cm (1).	1+1+1+1+1
	OR: Draw AB = 4 cm (1); Raise perpendiculars at A and B of length 4 cm (1); Join the ends to complete the square (1); Perimeter $= 16$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1
20	Draw AB = 7 cm (1); Draw perpendicular at B and mark BC = 5 cm (1); Draw perpendicular at A and mark AD = 5 cm, join D and C (1); Check that DC = 7 cm (1); Perimeter $= 2(7 + 5) = 24$ cm (1).	1+1+1+1+1
	OR: Draw AB = 5 cm (1); Raise perpendiculars at A and B of length 5 cm (1); Join the ends to complete the square (1); Perimeter $= 20$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.

- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 6	Half perimeter minus length.
2	**(a)** 20	Four equal sides.
3	**(c)** 18	Diameter $= 2\times$ radius.
4	**(d)** 11	Radius $=$ diameter $\div 2$.
5	**(d)** diameter	Property
6	**(c)** 15	Opposite sides of a rectangle are equal.
7	**(d)** 9	Half perimeter minus length.
8	**(c)** 32	Four equal sides.
9	**(c)** 24	Diameter $= 2\times$ radius.
10	**(d)** 13	Radius $=$ diameter $\div 2$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A rectangle (1); Other sides are 11 cm and 7 cm (1).	1+1
12	All sides are 10 cm (1); Perimeter $= 4 \times 10 = 40$ cm (1).	1+1
13	Diameter $= 2 \times 14 = 28$ cm (1); Longest chord is the diameter, 28 cm (1).	1+1
14	A rectangle (1); Other sides are 13 cm and 9 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius $= 10$ cm (1); Chord 18 cm vs diameter 20 cm (1); No: the diameter is the longest chord (1).	1+1+1
16	Open the compass to 5 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 5 cm from O (1).	1+1+1
17	Perimeter $= 2(15 + 6) = 42$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1
	OR: Radius $= 6$ cm (1); Chord 11 cm vs diameter 12 cm (1); No: the diameter is the longest chord (1).	1+1+1
18	Open the compass to 6 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 6 cm from O (1).	1+1+1
	OR: Perimeter $= 2(18 + 10) = 56$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 8 cm (1); Draw perpendicular at B and mark BC = 3 cm (1); Draw perpendicular at A and mark AD = 3 cm, join D and C (1); Check that DC = 8 cm (1); Perimeter $= 2(8 + 3) = 22$ cm (1).	1+1+1+1+1
	OR: Draw AB = 6 cm (1); Raise perpendiculars at A and B of length 6 cm (1); Join the ends to complete the square (1); Perimeter $= 24$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1
20	Draw AB = 9 cm (1); Draw perpendicular at B and mark BC = 6 cm (1); Draw perpendicular at A and mark AD = 6 cm, join D and C (1); Check that DC = 9 cm (1); Perimeter $= 2(9 + 6) = 30$ cm (1).	1+1+1+1+1
	OR: Draw AB = 7 cm (1); Raise perpendiculars at A and B of length 7 cm (1); Join the ends to complete the square (1); Perimeter $= 28$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.

- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 4	Property
2	**(b)** 11	Opposite sides of a rectangle are equal.
3	**(a)** 7	Half perimeter minus length.
4	**(a)** 40	Four equal sides.
5	**(c)** 30	Diameter $= 2\times$ radius.
6	**(c)** 20	Radius $=$ diameter $\div 2$.
7	**(a)** square	Property
8	**(d)** 12	Half perimeter minus length.
9	**(d)** 52	Four equal sides.
10	**(d)** centre	Property

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	All sides are 12 cm (1); Perimeter $= 4 \times 12 = 48$ cm (1).	1+1
12	Diameter $= 2 \times 18 = 36$ cm (1); Longest chord is the diameter, 36 cm (1).	1+1
13	A rectangle (1); Other sides are 16 cm and 10 cm (1).	1+1
14	All sides are 15 cm (1); Perimeter $= 4 \times 15 = 60$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Radius $= 9$ cm (1); Chord 16 cm vs diameter 18 cm (1); No: the diameter is the longest chord (1).	1+1+1
16	Open the compass to 7 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 7 cm from O (1).	1+1+1
17	Perimeter $= 2(20 + 11) = 62$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1
	OR: Radius $= 12$ cm (1); Chord 20 cm vs diameter 24 cm (1); No: the diameter is the longest chord (1).	1+1+1
18	Open the compass to 8 cm using the ruler (1); Fix the pointer at the centre O (1); Turn the pencil to complete the circle; every point is 8 cm from O (1).	1+1+1
	OR: Perimeter $= 2(25 + 14) = 78$ cm (1); Two sides are equal to the length (1); Opposite sides of a rectangle are equal (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw AB = 5 cm (1); Draw perpendicular at B and mark BC = 4 cm (1); Draw perpendicular at A and mark AD = 4 cm, join D and C (1); Check that DC = 5 cm (1); Perimeter $= 2(5 + 4) = 18$ cm (1).	1+1+1+1+1
	OR: Draw AB = 8 cm (1); Raise perpendiculars at A and B of length 8 cm (1); Join the ends to complete the square (1); Perimeter $= 32$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1
20	Draw AB = 10 cm (1); Draw perpendicular at B and mark BC = 7 cm (1); Draw perpendicular at A and mark AD = 7 cm, join D and C (1); Check that DC = 10 cm (1); Perimeter $= 2(10 + 7) = 34$ cm (1).	1+1+1+1+1
	OR: Draw AB = 9 cm (1); Raise perpendiculars at A and B of length 9 cm (1); Join the ends to complete the square (1); Perimeter $= 36$ cm (1); Sum of angles $= 4 \times 90^\circ = 360^\circ$ (1).	1+1+1+1+1

Mistakes to avoid:

- Taking the radius to be twice the diameter; the diameter is twice the radius.
- Choosing a compass opening of half AB or less for a perpendicular bisector, so the arcs do not meet.
- Changing the compass opening between the arcs drawn from A and from B.
- Calling every chord a diameter; only a chord through the centre is a diameter.

- Forgetting that the diagonals of a rectangle are equal and bisect each other, but cut at right angles only in a square.

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